

Gaming ISO 20022 Payment Messages: A Temporal Adversarial Benchmark for Purpose Codes, Party Fields, and Message Paths

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Abstract—ISO 20022 gives payment controls structured data, but message creators still decide where names, addresses, and purpose codes appear. We study *message gaming*: edits to purpose codes, party fields, or the message path that lower a modelled alert probability while preserving the payment’s economic meaning. A seeded generator gives each debtor–creditor pair latent behaviour and emits pain.001 and pacs.008 records in time order with a controlled defect rate; a cost-bounded evader applies up to two of eleven manipulations. All 2,089,794 generated and counterfactual records across ten runs are schema-valid and every gamed record passes its stated invariants. Evaluation is temporal: the earliest 60% of a run trains, the next 20% freezes alert thresholds, the latest 20% is scored. Screening-evasion lift (SEL) is a paired counterfactual on each manipulated payment. At a frozen 1% alert threshold, where at most 22.4% of gamed messages can be recovered, the best supervised model recovers 18.0% and falls to chance on a withheld misfielding family; the highest-mean-AUC gaming-label-free detector recovers 3.3%. A pre-specified confirmatory test finds no conclusive evidence that either the P1/P2/P3 family or an internal-contradiction label predicts primitive-level recovery. What does separate the families is the evidence required: on withheld path and structure edits no message-level detector exceeds AUC 0.538, while a history-only control reaches 0.851. Splitting a payment raises per-message lift to 1.44 but lowers attempt-level lift below 1. Results are synthetic and relative to a published screening abstraction.

Index Terms—ISO 20022, payment data quality, purpose codes, sanctions screening, anti-money laundering, adversarial evaluation, synthetic benchmark

I. INTRODUCTION

A payment message is both an instruction and a compliance record. Names, addresses, amounts, and purpose information feed sanctions, fraud, and anti-money-laundering controls, and ISO 20022 makes that information easier to parse. Structure does not remove discretion, though. A message creator still chooses which optional fields to populate, which purpose code to declare, how much detail to place in free text, and sometimes which path a payment takes. Screening is selective by design: Swift’s guiding principles identify the elements that normally warrant direct screening and let other fields provide context [1]. Market-practice guidance records that purpose codes are often misunderstood or placed in the wrong field [2]. The structured-address requirement that CBPR+ had scheduled

for 14 November 2026 was deferred by Swift in August 2026 after a community request, with revised timing under consultation [3]. CPPI’s harmonised requirements remain guidance rather than a mandate [4]. Unstructured and half-structured data will therefore coexist with structured data for years, and an ordinary data-quality error and a deliberate low-scrutiny choice can leave the same evidence.

That overlap creates an evaluation problem. A detector can look good by finding malformed records, even though malformed records are mostly benign. It can earn a high AUC by learning the edit signatures of one generator, then fail on a manipulation absent from training. And it can be evaluated with future information leaking into its history features. The question that matters operationally is narrower: at an alert capacity a payment operation could actually run, with thresholds frozen before the scored period, do deliberate changes remain distinguishable from routine error?

The contributions are eleven message-level manipulations with actors, costs, and stated invariants, checked on every gamed record; a generator with pair behaviour and time order; a temporal protocol with frozen thresholds and pair-clustered uncertainty; paired counterfactual SEL, which needs no separate control group; a pre-specified test of what predicts detectability; and a history-only positive control. Detectors are conventional on purpose. No real payment, customer, or institution data is used.

II. RELATED WORK AND GAP

Public AML resources model who pays whom, how much, and when: the Elliptic data set [5], AMLSim [6], AMLworld [7], SAML-D [8], and the recent AMLgentex and Tide generators with partial observability and configurable prevalence [9], [10]. None encodes the internal placement of ISO 20022 fields, so none can express a manipulation whose mechanism is that a name, code, or narrative sits in a different element. Adversarial machine learning on tabular financial data is the closest methodological line: imperceptible tabular perturbations [11], attacks on imbalanced fraud data [12], and constrained attacks and their benchmark [13], [14]. Those attacks move numeric features under a norm or constraint set; the edits here are

TABLE I

THE ELEVEN PRIMITIVES WITH ACTOR, COST $c(m)$, AND THE PRE-SPECIFIED CONTRADICTION LABEL (C: THE EDIT MAKES TWO ELEMENTS OF THE MESSAGE, OR AN ELEMENT AND THE CREDITOR'S PROFILE, INCONSISTENT).

ID	Operation (actor: originator or PSP unless stated)	c	C
P1a	move the ultimate-party name into remittance text	0.30	
P1b	true town and country into a free-text line; TwnNm set to another town	0.25	✓
P1c	truncate the listed party's name to four characters	0.15	
P1d	omit the ultimate debtor	0.20	
P2a	replace the purpose code by the lowest-weight plausible code	0.05	✓
P2b	replace the category purpose	0.08	✓
P2c	omit the purpose code	0.04	
P2d	declare a salary purpose over a goods or invoice narrative	0.10	✓
P3a	same single payment on the other path (actor with route choice)	0.45	
P3b	multiply the file's child count by 3 to 7, children sum to the amount	0.35	
P3c	an intermediary replaces the ultimate-party element	0.55	

discrete, semantics-preserving message operations whose cost is operational, in the classical cost-bounded evasion setting [15], [16]. Noisy payment-address parsing in real traffic [17] is the ordinary-error half of the same overlap. Transaction-purpose classification [18] enriches naturally occurring labels rather than detecting strategically chosen ones. The literature reviewed here offers no benchmark that combines message-level field operations with a controlled accidental-error floor and a temporal, capacity-constrained evaluation; this is a scoped observation, not a claim of exhaustive priority.

III. THREAT MODEL AND MANIPULATIONS

The adversary can construct or amend a payment instruction but controls neither the screening system nor its threshold. A black-box adversary knows public guidance and receives a coarse accept-or-hold outcome; its admissible set excludes the targeted code substitutions P2a, P2b, and P2d. A grey-box adversary also knows the relative ordering of the benchmark's purpose-code weights. Route choice (P3a) is a per-record capability held by a PSP, bank-side actor, or multi-channel originator, drawn for 30% of illicit records. Let x be an illicit instruction, m an admissible modification with operational cost $c(m)$, and $\pi(x)$ the alert probability in the published screening abstraction. The evader chooses $m^* = \arg \min_m \{\pi(m(x)) + \lambda c(m)\}$ subject to $V(m(x))$ and $|m| \leq k$, sampling among improving subsets with a softmax of temperature T . V is checked on every record. Amount, currency, debtor and creditor countries, and the economic originator and beneficiary are unchanged by every primitive. P3a changes the path of a single-transaction payment, P3b its decomposition into children that sum to the amount, and P3c the represented party chain, where an intermediary replaces the ultimate-party element. Primitives that need an element the record lacks are inadmissible for that record.

Table I lists the primitives. Whether the P1/P2/P3 family or the contradiction label predicts detectability was specified as a confirmatory question, with labels and tests, after the exploratory evaluation of an earlier version and before the corrected full-scale run (Section V-D).

IV. BENCHMARK AND PROTOCOL

A. Generator

A run draws $n/20$ debtor-creditor pairs with volume weights from a Dirichlet distribution. Each pair has a creditor industry and a purpose profile drawn around the industry's profile, a preferred path (pain.001 versus pacs.008), and a batching habit with a typical file size. It also has a party-chain pattern (the ultimate names it uses and how often one appears), an amount scale, and fixed names and addresses. Messages are emitted in time order by sampling pairs by weight; each carries its index as timestamp. A fraction η receives one ordinary defect (purpose text in remittance, partial address destructuring, absent ultimate party, absent category purpose, in the mix 35/30/20/15%). A fraction ρ is illicit: the listed party is the ultimate debtor or the debtor, and the evader edits the record. Benign remittance text and party names have variable length, so no edit leaves a length that benign traffic does not show; an audit checks this on every run. Records serialise to pain.001.001.09 or pacs.008.001.08. Every generated record and every counterfactual original of the ten main runs was validated against the official XSDs (2,089,794 of 2,089,794 valid), and every gamed record passed its invariant check (0 violations in 89,794 records). Full distributions are in the supplement.

B. Screening abstraction

A name leg matches listed names against the elements the current path screens: debtor, creditor, and ultimate-party names on pain.001, debtor and creditor names only on the interbank leg. For an illicit record whose listed name sits in a screened element it alerts with probability P_{name} , reduced to $0.39P_{\text{name}}$ when the name is truncated. A score leg combines purpose weight w_p (0.08 to 0.45), corridor weight w_c (0.05 to 0.40), amount, and completeness,

$$s = w_p + w_c + 0.2 \min(1, a/(n_{\text{ch}} \cdot 250,000)) + 0.10u + 0.08t, \quad (1)$$

where a is the payment amount, n_{ch} the number of children, u indicates a missing ultimate debtor and t a missing town or country; only the amount term is diluted by splitting. The leg alerts with probability $p_{\text{score}} = 1/(1 + e^{-8(s-\tau)})$. A residual floor $f = 0.10$ represents controls outside the evader's reach. Per message, $\pi = 1 - (1 - p_{\text{name}})(1 - p_{\text{score}})(1 - f)$; per attempt, the name leg and floor act once and the score leg once per child, so $\pi_{\text{att}} = 1 - (1 - p_{\text{name}})(1 - f)(1 - p_{\text{score}})^{n_{\text{ch}}}$. This models perfectly correlated name-leg and floor outcomes across children and a separate score-leg draw per child; the fully independent extreme, $1 - (1 - \pi)^{n_{\text{ch}}}$, is reported in the supplement. The evader optimises the per-message form. Baseline survival $S = (1 - \pi_0)/(1 - f)$ is split between the legs by a name share σ : $P_{\text{name}} = (1 - S^\sigma)/q$, where q is the share of unevaded illicit probe records whose listed name sits in a screened element, and τ is bisected so that the probe's mean score-leg probability equals $1 - S^{1-\sigma}$. The main setting $\pi_0 = 0.45$, $\sigma = 0.6$ solves to $P_{\text{name}} = 0.337$, $\tau = 0.782$. Weights are illustrative constants, not any institution's rule set.

C. Features, detectors, protocol

Message-level features are the consistency checks C1 (code not in list), C2 (town or country missing, address token in free text), C3 (purpose implausible for the creditor’s industry, ultimate debtor absent), and C4 (salary code over a goods narrative). Completeness and length auxiliaries complete the vector. History features use only the pair’s strictly earlier messages: purpose novelty (C5), path novelty, split-count deviation, ultimate-party novelty and presence deviation, and amount deviation. No label enters any feature. D_1 is the fixed rule $C1 + 0.5 C2 + C3 + 1.5 C4$. D_2 is a gradient-boosted classifier (scikit-learn `HistGradientBoostingClassifier`, 150 iterations, learning rate 0.1, 31 leaves, other settings default) and D_6 an ℓ_2 logistic regression (`LogisticRegression`, $C = 1$, lbfgs, 1,000 iterations, standardised inputs) on one-hot codes, corridor, amount, and completeness. Both train on gaming labels. D_5 is a local outlier factor on the message-level features of benign training rows [19]; it and D_7 use no gaming label but do assume a trusted benign training stream, which we call gaming-label-free. D_3 (isolation forest) and D_4 (rules plus forest) are in the supplement. D_7 is the history-only positive control, the sum of the six history deviations standardised with training-window means and standard deviations only. No hyperparameter is tuned on the benchmark.

Each run has $n = 200,000$ records with $\eta = 0.15$, $\rho = 0.05$, $k = 2$, $\lambda = 0.18$, $T = 0.10$, and $f = 0.10$. The earliest 60% trains every learned detector. On the next 20% the score threshold for each alert capacity (1% and 0.1%) is frozen, with a label-blind fraction for tied scores. The latest 20% is the test window on which every number below is computed. An illicit record for which no admissible subset lowers the objective, or for which the softmax draws the no-edit option, stays unedited; it is not in the positive class and remains in the scored population as an unevaded illicit record (0.5% of the test window). Gamed records are therefore 4.5% of the test window rather than ρ , so no detector can recover more than 22.4% at 1% capacity. We report AUC, average precision (AP), and recovery and precision at the frozen threshold, as means over ten seeds with between-seed standard deviations, and pair-clustered bootstrap intervals within each test window. SEL is paired. Each gamed test record contributes its original x and edited $m(x)$, both scored by the same detector with the same prior history, and $SEL = \sum(1 - \pi_m) / \sum(1 - \pi_0)$. Residual SEL multiplies each term by one minus the frozen alert decision. Leave-one-family-out retrain D_2 and D_6 without one family (at $k = 1$, pure-family records). Splitting’s attempt-level lift is 0.94 under the main dependence assumption and 0.49 under the independent extreme (supplement).

V. RESULTS

A. Recovery at a frozen threshold

Table II separates ranking from operational recovery under a threshold frozen before the scored period. Gamed records are 4.5% of the test window, so 1% capacity can recover at most 22.4%. The fixed rules D_1 recover 11.2% at precision 0.499:

TABLE II

TEST-WINDOW RESULTS, MEAN $\pm$ BETWEEN-SEED S.D. OVER TEN RUNS OF 200,000 RECORDS. REC. AND PREC. ARE AT THE THRESHOLD FROZEN ON THE VALIDATION WINDOW AT 1% CAPACITY; THE WIDEST PAIR-CLUSTERED 95% HALF-INTERVAL FOR REC. IS 0.020. [†] TRAINS ON GAMING LABELS.

Detector	AUC	AP	Rec.@1%	Prec.@1%	Alerts/hit	Rec.@0.1%
D_1 rules	0.771 $\pm$ 0.004	0.321	0.112 $\pm$ 0.005	0.499	2.0	0.022
D_5 LOF	0.775 $\pm$ 0.005	0.249	0.033 $\pm$ 0.005	0.136	7.5	0.012
D_7 history	0.717 $\pm$ 0.006	0.123	0.035 $\pm$ 0.004	0.205	4.9	0.003
D_2 GBM [†]	0.886 $\pm$ 0.008	0.498	0.180 $\pm$ 0.014	0.818	1.2	0.021
D_6 logistic [†]	0.862 $\pm$ 0.009	0.436	0.155 $\pm$ 0.013	0.712	1.4	0.020
Capacity ceiling	–	–	0.224	1.000	–	0.022

TABLE III

LEFT: LEAVE-ONE-FAMILY-OUT AUC ON PURE-FAMILY TEST RECORDS ($k = 1$, TEN SEEDS; IN $\rightarrow$ OUT FOR THE RETRAINED SUPERVISED MODELS, SINGLE VALUE FOR GAMING-LABEL-FREE ONES). RIGHT: GENERATOR SHIFT, DETECTORS FITTED AND THRESHOLDS FROZEN ON CONFIGURATION A, SCORED ON B’S TEST WINDOW (THREE SEEDS); RECOVERY AND ALERT RATE ARE UNDER A’S FROZEN THRESHOLD, NOT AT A 1% ALERT RATE ON B; B $\rightarrow$ B IN PARENTHESES.

	P1	P2	P3	AUC A $\rightarrow$ B	Rec. (A thr.)	Alerts
D_1	0.551	0.825	0.383	0.730 (0.730)	0.139	0.017
D_5	0.514	0.834	0.538	0.765 (0.786)	0.019	0.006
D_7	0.576	0.715	0.851	0.697 (0.702)	0.035	0.007
D_2	0.633 $\rightarrow$ 0.490	0.976 $\rightarrow$ 0.641	0.550 $\rightarrow$ 0.502	0.875 (0.897)	0.375	0.045
D_6	0.591 $\rightarrow$ 0.482	0.964 $\rightarrow$ 0.677	0.543 $\rightarrow$ 0.525	0.869 (0.879)	0.336	0.034

one alert in two is a gamed message. The local outlier factor D_5 has the best gaming-label-free AUC (0.775 against 0.771 for D_1) and the second-best AP, yet the lowest gaming-label-free recovery, 3.3%; the history-only control D_7 recovers 3.5%. Among gaming-label-free detectors neither AUC nor AP orders recovery at the frozen threshold. The supervised D_2 reaches AUC 0.886 and recovers 18.0% (80% of the ceiling, precision 0.818); logistic regression on the same inputs recovers 15.5%. Realised alert rates on the test window stay within 0.002 of the 1% target, so the frozen thresholds transferred across the window boundary. The exact top-1% ranking metric differs from frozen-threshold recovery by at most 0.011 (supplement). Between-seed standard deviations are at most 0.009 for AUC and 0.014 for recovery.

B. Generalisation, history, and shift

Table III (left) retrains the supervised models without one family and scores pure-family test records. With P1 withheld, D_2 falls from 0.633 to 0.490 (between-seed s.d. 0.018), and no gaming-label-free detector exceeds 0.598 on P1, whose records are dominated by name truncation. The inversion behind the low supervised score is systematic. In the reduced training window an absent ultimate debtor is more common among benign records (27%) than among the remaining gamed ones, and the gamed rate given absence is 1.1% against 2.6% given presence. The two P1 edits that remove the ultimate debtor therefore score 0.222 (P1a) and 0.210 (P1d) against benign traffic, while truncation, which leaves it in place, scores 0.568. With P2 withheld the supervised drop is 0.976 $\rightarrow$ 0.641 while the gaming-label-free D_1 and D_5 stay at 0.825 and 0.834.

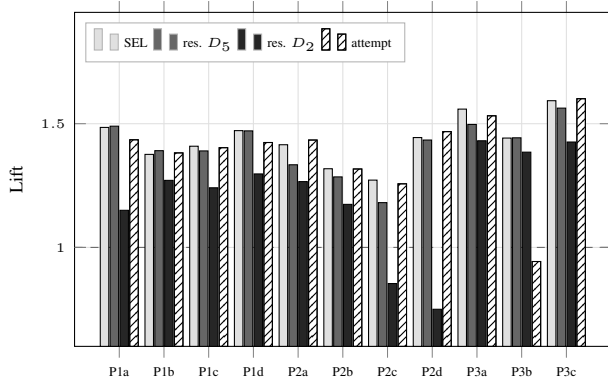


Fig. 1. Paired SEL per primitive on the test window (ten seeds): modelled lift, residual lift after D_5 and D_2 at the frozen 1% threshold, and attempt-level lift when every child transaction is screened; residual lift after D_1 and D_7 is in the supplement.

P3 is the pre-specified contrast. No message-level detector exceeds AUC 0.538 (0.383 to 0.538; the withheld D_2 0.502), and the history-only D_7 reaches 0.851. Path substitution, decomposition, and party-chain changes leave the message coherent but break the pair’s prior pattern. Table III (right) fits the detectors and freezes thresholds on configuration A, then scores configuration B, whose pair behaviour, industry and corridor mix, defect composition, batching, and string lengths all differ. Rankings transfer: every AUC is within 0.022 of its B→B value. Thresholds do not: D_2 ’s A-frozen threshold alerts 4.5% of B’s test window instead of 1%, which is why its recovery rises to 37.5% at precision 0.365, while D_5 ’s alerts 0.6% and recovers 1.9%. A capacity budget therefore needs recalibration on the target traffic even when the ranking model does not. At 0.5% prevalence (supplement), precision at the frozen threshold falls to 0.080 for D_2 and 0.114 for D_1 , so the fixed rules overtake the supervised model, at 13 and 9 alerts per detected manipulation.

C. Paired screening-evasion lift

Fig. 1 gives paired SEL per primitive. Manipulation raises modelled pass-through by 27–59% per primitive and 40% overall, with pair-clustered half-intervals of at most 0.122. Residual lift shows what each detector removes at its frozen threshold. D_2 pushes the code–narrative mismatch P2d to 0.75 and the omitted code P2c to 0.85, both below 1. It does so by alerting 48.3% and 32.6% of them while alerting under 1% of their unmanipulated originals. The fixed rules bring P2d to 0.96. The path and party-chain edits keep most of their lift under every detector: the smallest residual lifts for P3a and P3c are 1.43 and 1.43. The history-only D_7 is the detector that reaches P3c (10.5% against 0.8% for D_1). Splitting shows the value of the attempt-level view. P3b’s per-message lift is 1.44, but once every child transaction is screened its attempt-level lift is 0.94, below 1, so file decomposition pays only against per-message thresholds. For every other primitive the two forms agree within 0.05.

D. What predicts detectability

The reviewed version of this work read its results as showing that detectability follows whether an edit leaves a checkable contradiction rather than its family. We specified that reading as a confirmatory hypothesis, with labels and tests, after that exploratory evaluation and before the first run of the corrected protocol (Table I; supplement). The contradiction label has almost no explanatory value; the family effect is suggestive but inconclusive with eleven primitives. For the supervised D_2 the contradiction label explains $R^2 = 0.04$ of the variance in per-primitive recovery at the frozen threshold (exact permutation $p = 0.567$ over all 330 relabellings; contradiction-bearing primitives 19.8% against 14.8% for coherent ones). The family explains $R^2 = 0.35$ ($p = 0.125$ over 11,550). For the gaming-label-free D_5 the figures are 0.00 ($p = 0.955$) and 0.49 ($p = 0.066$). Cost correlates negatively with recovery ($\rho = -0.60$, $p = 0.06$); frequency does not. In leave-one-primitive-out prediction the contradiction grouping is worse than the grand mean and the family grouping improves on it only marginally (MAE 0.123 and 0.099 against 0.107). The per-primitive rates show why. The omitted purpose code P2c, labelled coherent, is the second most recovered primitive (32.6%), because absence is itself a feature. The address contradiction P1b, labelled contradiction-bearing, is barely recovered (8.6%), because no feature compares the structured town with the free-text line. The case analysis suggests, without testing it in the same way, that detection depends on whether the detector observes the evidence source the primitive modifies; a contradiction that no feature reads goes unseen. An exploratory grouping by evidence source is in the supplement. What does separate the families is the kind of evidence required, as the leave-one-family-out result shows: P3 edits are separated by prior pair history and by no tested message-level detector at a useful level. Of the two directional expectations recorded before the rerun, one held: P1 and P3 behave alike for message-level detectors when withheld. One did not: P2 is the easiest family for fixed rules, not the hardest.

VI. DISCUSSION, LIMITATIONS, AND CONCLUSION

Three results survive every protocol correction. Ranking quality does not translate into recovery at a frozen threshold: gaming-label-free detectors with AUC between 0.717 and 0.775 recover between 3.3% and 11.2% of gamed messages when 1% of traffic may be alerted, against a ceiling of 22.4%. The highest-mean-AUC detector has the lowest mean recovery. A supervised model that recovers 18.0% in-distribution falls to chance on a family it has not seen, and the reason can be read off the training distribution: the ordinary-error floor makes the signature of the unseen edit look benign. The primitives that reroute, decompose, or re-chain a payment are not usefully separated by the tested message-level checks and are visible to prior pair history. The control that addresses them is longitudinal, not a better validator. The claim drawn from the earlier evaluation, that an internal contradiction predicts detectability, did not survive its confirmatory test; the case analysis suggests that what matters is whether a detector reads

the field the edit touches. Under a generator shift the rankings hold while frozen thresholds do not, and at low prevalence precision collapses for every detector.

Limitations. The screening abstraction is transparent, not a reconstruction of any engine, so SEL is a stress measure relative to it. Its weights, the defect mix, pair behaviour, and illicit prevalence are assumptions. The SEL range across the calibration grid (1.14 to 2.85) shows how much the modelled threat depends on them. Generator and detectors share a feature vocabulary; the audit finds 0.0% of benign and 1.2% of gamed test records outside the benign training ranges and at most 5% of any detector’s alert mass on them. Domain ratings of the primitives have not been collected; the questionnaire and blinded procedure are released and the pre-specified test reruns unchanged on raters’ labels. XSD validity is not scheme or business acceptance, and time order is synthetic and without drift.

Artifact

The generator, harness, result files, and verification scripts are public at https://github.com/abhisheksharma2411/ISO-20022-purpose-code_gaming_benchmark (MIT) and archived at <https://doi.org/10.5281/zenodo.23049366>. Every result file records the commit that produced it (17bb868dd2b1). One command regenerates every table and figure, and one command validates every message against the XSDs and every gamed record against its invariants.

REFERENCES

- [1] Swift, *Guiding Principles for Screening ISO 20022 Payments Messages*, Oct. 2021.
- [2] Payments Market Practice Group, *ISO 20022 Market Practice Guidance: Regulatory Reporting, Purpose of Payment and Category Purpose*, ver. 1.1, Jul. 2024.
- [3] Swift, “Swift accepts community request to extend structured address migration in ISO 20022 payment messages,” 27 Aug. 2026. [Online]. Available: <https://www.swift.com/news-events/news/swift-accepts-community-request-extend-structured-address-migration-iso-20022-payment-messages>
- [4] Committee on Payments and Market Infrastructures, *Harmonised ISO 20022 Data Requirements for Enhancing Cross-Border Payments*, Bank for International Settlements, updated report, Feb. 2026.
- [5] M. Weber *et al.*, “Anti-money laundering in Bitcoin: Experimenting with graph convolutional networks for financial forensics,” in *KDD Workshop on Anomaly Detection in Finance*, 2019, arXiv:1908.02591.
- [6] T. Suzumura and H. Kanezashi, “Anti-money laundering datasets: InPlusLab anti-money laundering datasets,” 2021. [Online]. Available: <https://github.com/IBM/AMLSim>
- [7] E. Altman *et al.*, “Realistic synthetic financial transactions for anti-money laundering models,” in *Advances in Neural Information Processing Systems*, vol. 36, 2023.
- [8] B. Oztas *et al.*, “Enhancing anti-money laundering: Development of a synthetic transaction monitoring dataset,” in *Proc. IEEE ICEBE*, 2023, pp. 47–54.
- [9] J. Östman *et al.*, “AMLgentex: Mobilizing data-driven research to combat money laundering,” arXiv:2506.13989, 2025.
- [10] M. van den Beukel, J. M. Rožanec, and A.-L. Varbanescu, “Tide: A customisable dataset generator for anti-money laundering research,” arXiv:2603.01863, 2026.
- [11] V. Ballet *et al.*, “Imperceptible adversarial attacks on tabular data,” arXiv:1911.03274, 2019.
- [12] F. Cartella *et al.*, “Adversarial attacks for tabular data: Application to fraud detection and imbalanced data,” in *Proc. AAAI Workshop on Artificial Intelligence Safety (SafeAI)*, 2021, arXiv:2101.08030.
- [13] T. Simonetto, S. Dyrnishi, S. Ghamizi, M. Cordy, and Y. Le Traon, “A unified framework for adversarial attack and defense in constrained feature space,” in *Proc. IJCAI*, 2022, pp. 1313–1319.
- [14] T. Simonetto, S. Ghamizi, and M. Cordy, “TabularBench: Benchmarking adversarial robustness for tabular deep learning in real-world use-cases,” in *Advances in Neural Information Processing Systems*, vol. 37, Datasets and Benchmarks Track, 2024.
- [15] M. Brückner and T. Scheffer, “Stackelberg games for adversarial prediction problems,” in *Proc. ACM SIGKDD*, 2011, pp. 547–555.
- [16] B. Biggio *et al.*, “Evasion attacks against machine learning at test time,” in *Proc. ECML PKDD*, 2013, pp. 387–402.
- [17] H. Hammami, L. Baligand, and B. Petrovski, “Fighting crime with Transformers: Empirical analysis of address parsing methods in payment data,” in *Proc. NAACL-HLT Industry Track*, 2024, pp. 201–212.
- [18] A. J. G. Busson *et al.*, “Hierarchical classification of financial transactions through context-fusion of transformer-based embeddings and taxonomy-aware attention layer,” arXiv:2312.07730, 2023.
- [19] M. M. Breunig, H.-P. Kriegel, R. T. Ng, and J. Sander, “LOF: Identifying density-based local outliers,” in *Proc. ACM SIGMOD*, 2000, pp. 93–104.